

Call for Expressions of Interest

<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	Mapping the design space of service robot external human-machine interfaces (eHMIs)
<i>Company:</i>	n/a

Contact Details

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Additional Info

<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☐
<i>Custom hardware required:</i>	☐
	If Applicable: Please provide details
<i>Relevant research areas (if applicable)¹:</i>	Click or tap here to enter text.

Project Background Information

Service robots are starting to be deployed in technology trials and commercial operations across various publicly accessible places – shopping centres, restaurants, public streets, airports, etc. But one problem is that often such robots take actions that are illegible to co-present people who are not “primary users” of the robots. At present our understanding of the different ways that such robots display their “intentions” to people nearby is limited. For instance, delivery robots operating in various cities across the world tend to adopt vehicle-style indicator lights for left and right turns, red lights for braking, etc., but it is not clear that this is the best external human-machine interface (eHMI) to provide legibility on the streets.

This project aims to explore the general problem of eHMIs for mobile robots. The project will map out in experimental scenarios the features of different strategies for eHMIs, which can take a multimodal form (e.g., voice announcements, lights, text displays, etc.).

¹ An overview of different research areas in the school of computer science can be found [here](#).

This project will take place in the Cobot Maker Space, using a Turtlebot as the platform to explore the design space of eHMIs for mobile robots that are around passers-by, bystanders, and co-present people in general.

Requirements

The group must provide:

A working Turtlebot based experimental framework that is repeatable and reconfigurable. This will involve implementing basic wayfinding and navigation with a Turtlebot, and a physical setup within the CMS that can be used in lab-based experimentation.

A set of proposed and implemented example eHMIs to test using various modalities (to be discussed). This will involve finding ways to display, project or otherwise present signalling that is visible, audible, etc. on or next to a Turtlebot that is undertaking wayfinding and navigation actions.

Analysis of the results of experiments run with human participants in a distinct set of scenarios (to be discussed) that test eHMIs on the Turtlebot.

Presentation of results and practical recommendations for robot designers who are wanting to support legible robot actions.

Skills

Highly desirable:

Experience with Turtlebots platform
Experience with ROS

Desirable:

Interest in robotics
Interest in planning and conducting lab-based user trials with robots

Submission / Award information

<i>Date of Submission:</i>	September 2026
<i>Date of Pitch:</i>	October 2026
<i>Notification of award:</i>	October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.